

The Quantum Ground

Sequenced Excitations, Information Emergence, and Cross-Scale Coherence

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Overview

The Quantum Ground framework extends the Imprint Hypothesis, Detector-Plane Imaging (DPI), and Quantum Measurement Stack (QMS) by providing a unified observational vocabulary spanning quantum, microscopic, macroscopic, and cosmological scales.

The framework introduces **no new physical laws**.

Instead, it proposes a consistent way of describing reality from quantum excitations to emergent structures through the concepts of:

- Quantum Ground
 - Localized Excitation Events
 - Contextual Imprints
 - Sequenced Excitations
 - Cross-Scale Coherence
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Core Proposition

Physical reality may be viewed as an inseparable combination of:

$$Q_G = (\text{Spacetime, Quantum Fields, Energy})$$

where:

Spacetime

Provides geometry and causal structure.

Quantum Fields

Provide identity and interaction channels.

Energy

Provides excitation potential.

The Quantum Ground is not a new field or hidden variable.

It is a unified description of structures already present in modern physics.

Localized Excitation Events

A quantum particle is defined as:

A localized, identity-bearing excitation of a quantum field occurring within spacetime.

Examples:

- Electron-field excitation → Electron
- Electromagnetic-field excitation → Photon
- Quark-field excitation → Quark

Particle identity is inherited directly from the excited field.

Contextual Imprints

The Imprint Hypothesis proposes that wave-like phenomena arise from interaction history rather than representing a separate ontology.

Conceptual Sequence

Localized Excitation Event

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Interaction Dynamics

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Contextual Imprint

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Measurement

Interference, diffraction, tunneling, and coherence phenomena are interpreted as contextual imprints generated through particle-field interactions.

Sequenced Excitations

Central Principle

Information is preserved sequencing.

Information emerges from the ordered relationships between excitation events occurring within the Quantum Ground.

The same physical ingredients may produce radically different outcomes depending on how excitation events are organized and preserved through time.

Examples include:

- Quantum coherence
- Atomic structure
- Molecular chemistry
- Biological organization
- Macroscopic systems

Complexity emerges through sequencing rather than through additional physical ingredients.

Cross-Scale Coherence

Principle

A valid ontology should remain self-consistent across all observable scales.

Example:

Table

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Material Structure

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Molecular Organization

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Atomic Configuration

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Quantum Particles

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Field Excitations

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Quantum Ground

These are not different realities.

They are different resolutions of the same physical system.

Measurement and Detector-Plane Causality

The Quantum Measurement Stack identifies the detector plane as the effective measurement boundary.

Measurement Flow

Quantum Ground

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Localized Excitation

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Contextual Imprint

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Detector-Plane Registration

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Measurement Record

Information discarded at the detector plane cannot be recovered through later processing.

Measurement therefore reflects detector-plane causality rather than observer-dependent effects.

Relation to Previous Work

Imprint Hypothesis

Established:

- Field primacy
- Particle realism
- Wave contextuality

Detector-Plane Imaging (DPI)

Established:

- Measurement-plane coherence evolution
- Pre-discretization coherence diagnostics

Quantum Measurement Stack (QMS)

Established:

- Detector-plane causality
- Measurement integrity benchmarking
- Localization of irreversible information loss

Quantum Ground Framework

Introduces:

- Sequenced Excitations
 - Information as Preserved Sequencing
 - Cross-Scale Coherence
 - Unified Observational Ontology
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Key Takeaway

The Quantum Ground framework proposes that:

- Fields are persistent.
- Particles are localized excitations.

- Wave behavior reflects contextual imprints.
- Information emerges through sequenced excitations.
- Measurement fixes surviving imprints.
- Macroscopic reality emerges through cross-scale coherence.

The framework remains fully compatible with Quantum Field Theory while providing a unified vocabulary connecting quantum phenomena, information, measurement, and emergent reality.

Project Resources

Publications

Imprint Hypothesis <https://zenodo.org/records/20470449>

Detector-Plane Imaging (DPI) <https://zenodo.org/records/18994419>

Quantum Measurement Stack (QMS) <https://zenodo.org/records/19153014>

QMCTB Benchmark Artifacts <https://zenodo.org/records/19147730>

Code Repository

<https://github.com/srikarr20>

Summary Statement

Reality may be viewed as localized excitation events occurring within a common Quantum Ground, where information emerges through the sequencing of those events and observable structure arises through cross-scale coherence.